

CLINICAL PROGRAMMING CASE STUDY

Treatment-Emergent Adverse Event Analysis

Sanofi EFC10261 | Phase II Non-Small Cell Lung Cancer Trial

Portfolio	Oncology Clinical Programming Portfolio
Milestone	5 - End-to-End Clinical Safety Programming
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Executive Summary

This report presents an end-to-end treatment-emergent adverse event (TEAE) analysis using clinical trial data from Sanofi study EFC10261, a Phase II study in non-small cell lung cancer. The workflow begins with assessment of the SDTM Adverse Events (AE) domain, integrates subject-level treatment and population information from the Milestone 4 ADSL-style dataset, derives an ADAE-style analysis dataset, and produces safety tables and a figure suitable for clinical review.

The safety population included 453 subjects. Among them, 420 subjects (92.7%) experienced at least one TEAE, contributing 8,743 treatment-emergent event records. A total of 205 subjects (45.3%) had a maximum Grade 3 or Grade 4 TEAE, and 138 subjects (30.5%) experienced at least one serious TEAE. Alopecia was the most common Preferred Term, affecting 141 subjects (31.1%).

Key takeaway: The analysis demonstrates a reproducible clinical programming workflow that preserves traceability from SDTM source records through analysis-ready derivations, population logic, quality control, and employer-facing safety outputs.

1. Study Context and Report Objective

The EFC10261 submission package provides publicly available patient-level SDTM data from a Phase II oncology trial in non-small cell lung cancer. Milestone 5 uses these data to simulate core responsibilities of a clinical statistical programmer: understanding source data, applying analysis definitions, creating analysis-ready datasets, validating derivations, and communicating results.

The objective of this case study was to evaluate treatment-emergent adverse events in the safety population and summarize their incidence, frequency, toxicity grade, and seriousness. This report documents the analysis approach and interprets the resulting outputs; the associated README remains a concise guide to the repository structure and programs.

2. Data Sources and Analysis Population

Dataset	Level	Role in analysis
SDTM AE	One record per adverse event	Source event terminology, timing, toxicity grade, seriousness, relationship, action, and outcome
Milestone 4 ADSL	One record per subject	Treatment dates, demographics, disposition information, and safety population flag

The safety population was defined using SAFETY == "Y" in the Milestone 4 ADSL-style dataset. This resulted in a fixed denominator of 453 unique subjects. All subject-incidence percentages in the TEAE summaries and figure use this denominator, including subjects who did not experience a TEAE.

3. Programming and Analysis Methods

3.1 Source Data Assessment

Before deriving analysis variables, the SDTM AE domain was assessed for structure, terminology, timing, and completeness. Reviews covered Preferred Term and System Organ Class terminology, AE start and end study days, toxicity grade, seriousness, treatment relationship, action taken, outcome, event pattern, and source treatment-period variables. These checks established the values actually available in the study and prevented assumptions about standard variables that were not present.

For example, the source data do not contain AESEV. Toxicity intensity is represented by numeric AETOXGR values ranging from Grade 1 through Grade 4. Seriousness is represented separately by AESER with values Y and N. This distinction was preserved throughout the analysis.

3.2 ADAE-Style Dataset Construction

The event-level AE records were merged with selected subject-level variables from ADSL using RUSUBJID. The resulting ADAE-style dataset preserves the original AE records while adding treatment dates, population membership, and analysis flags required for downstream summaries.

Derived variable	Purpose
ONTRTFL	Identifies events classified within the on-treatment period
POSTTRTFL	Identifies events classified after the treatment period
TRTEMFL	Identifies treatment-emergent adverse events used in the primary safety analyses

Treatment-emergent analyses were restricted to safety subjects with TRTEMFL == "Y". Source records that were non-treatment-emergent or could not be classified were retained in the ADAE-style dataset and documented in validation listings rather than silently removed.

3.3 TEAE Incidence and Event Frequency

The primary TEAE summary was produced at three levels: overall, System Organ Class (SOC), and Preferred Term (PT) within SOC. Two distinct measures were retained:

- Subject incidence: the number and percentage of unique safety subjects experiencing at least one qualifying event in a category.
- Event frequency: the total number of qualifying adverse-event records in the category.

A subject is counted once in a subject-incidence row regardless of the number of qualifying records for that category. Every qualifying record contributes to event frequency. Because subjects can experience events in multiple SOC and PTs, counts across categories are not additive.

3.4 Toxicity Grade and Seriousness

TEAE records were collapsed to one record per safety subject to derive whether the subject experienced any TEAE, the maximum observed toxicity grade, and whether at least one TEAE was serious. Subjects without a TEAE retained a missing maximum toxicity grade. Grade 0 was not assigned because it was not

an observed source value and would imply a recorded toxicity grade rather than the absence of an event.

4. Results

4.1 Overall TEAE Burden

Measure	Result
Safety population	453 subjects
Subjects with at least one TEAE	420 (92.7%)
Subjects without a TEAE	33 (7.3%)
Treatment-emergent event records	8,743

TEAEs were reported for 420 of the 453 safety subjects, indicating that treatment-emergent events were common in this oncology study population. The difference between 8,743 event records and 420 affected subjects also shows the importance of separating event frequency from subject incidence: many subjects experienced multiple qualifying events.

4.2 Most Frequently Affected System Organ Classes

System Organ Class	Subjects	Incidence	Events
General disorders and administration site conditions	259	57.2%	1,514
Gastrointestinal disorders	220	48.6%	1,013
Respiratory, thoracic and mediastinal disorders	209	46.1%	1,023
Skin and subcutaneous tissue disorders	196	43.3%	1,787
Nervous system disorders	159	35.1%	1,012

General disorders and administration site conditions affected the greatest number of subjects. Skin and subcutaneous tissue disorders had the largest event count among the five leading SOCs, illustrating that the category with the greatest event frequency is not necessarily the category affecting the greatest number of unique subjects.

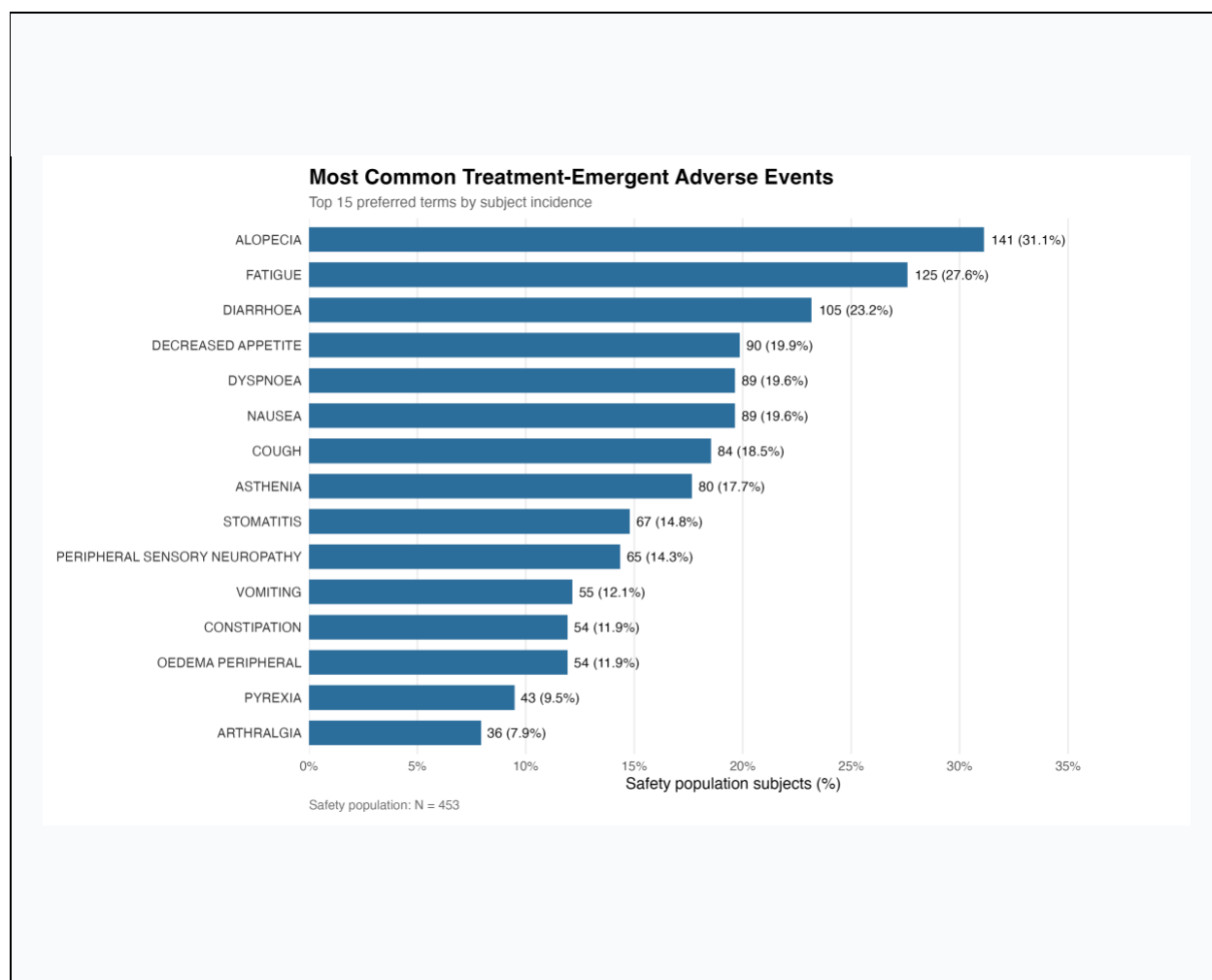
4.3 Most Frequently Reported Preferred Terms

Preferred Term	Subjects	Incidence
Alopecia	141	31.1%
Fatigue	125	27.6%
Diarrhoea	105	23.2%
Decreased appetite	90	19.9%
Dyspnoea	89	19.6%
Nausea	89	19.6%

Alopecia was the most common TEAE Preferred Term, affecting nearly one-third of the safety population. Fatigue, diarrhoea, decreased appetite, dyspnoea, and nausea were also among the most

frequently reported events. The complete analysis output includes 383 PT rows nested beneath 22 SOC rows, in addition to the overall TEAE row.

Figure 1. Top 15 TEAE Preferred Terms by Subject Incidence



Note: Percentages use the safety population denominator (N = 453).

4.4 Maximum Toxicity Grade and Serious TEAEs

Category	Subjects, n (%)
Any TEAE	420 (92.7%)
Maximum toxicity Grade 1	46 (10.2%)
Maximum toxicity Grade 2	169 (37.3%)
Maximum toxicity Grade 3	164 (36.2%)
Maximum toxicity Grade 4	41 (9.1%)
Maximum toxicity Grade 3 or higher	205 (45.3%)
At least one serious TEAE	138 (30.5%)

The mutually exclusive maximum-grade categories total 420 subjects with at least one TEAE. Overall, 205 safety subjects (45.3%) experienced a maximum Grade 3 or Grade 4 TEAE. Serious TEAEs occurred in

138 subjects (30.5%). At the event level, 242 records were classified as serious; collapsing to the subject level shows that those records occurred among 138 unique subjects.

Interpretive note: Toxicity grade and seriousness are related but distinct clinical concepts. A serious event is defined by regulatory outcome criteria, while toxicity grade reflects intensity. The source data included serious Grade 1 events and non-serious Grade 4 events, confirming that one should not be used as a substitute for the other.

5. Validation and Quality Control

Quality-control checks were embedded directly in the R programs using `stopifnot()` statements and targeted data reviews. The workflow validated both structural assumptions and expected analysis results.

- The safety population contained 453 unique subjects.
- All ADAE subjects belonged to the subject-level source dataset.
- The TEAE population contained 420 unique subjects and 8,743 event records.
- Subject counts did not exceed the safety denominator, and percentages remained between 0% and 100%.
- Event counts were not smaller than their corresponding subject counts.
- The subject-level grade dataset contained exactly one record per safety subject.
- The four mutually exclusive maximum-grade categories totaled 420 subjects.
- The Grade 3-or-higher count equaled the combined Grade 3 and Grade 4 subject counts.
- The plotting dataset contained exactly 15 Preferred Terms with reproducible ordering and alphabetical handling of ties.

Analysis-ready outputs retained numeric count and percentage fields for QC and reuse. Separate display-ready outputs formatted the results as n (%) for reporting. This separation prevents presentation formatting from interfering with numeric validation.

6. Clinical Programming Interpretation

This case study is designed as a programming demonstration rather than a comparative treatment-effect analysis. The results describe the frequency and distribution of treatment-emergent adverse events within the available safety population. They should not be interpreted as establishing causality or comparing treatment arms.

Several programming decisions were especially important. First, the denominator remained fixed at all 453 safety subjects, including the 33 without a TEAE. Second, subject incidence was kept distinct from event frequency. Third, the terminology hierarchy was preserved by presenting PTs within SOCs. Finally, maximum grade and seriousness were derived at the subject level so that repeated events did not inflate subject counts.

Together, these decisions produce outputs that are clinically interpretable, reproducible, and traceable to the underlying source data.

7. Limitations

- The analysis is based on a publicly available, de-identified submission package and is intended for portfolio and educational use.
- The ADAE-style dataset demonstrates analysis-ready principles but is not presented as a fully specification-driven, CDISC-compliant production ADAE deliverable.
- The report summarizes one safety population and does not estimate between-arm treatment effects or infer causality.
- Treatment-emergent classification depends on the available study-day and treatment-period information; source records that could not be classified were retained and reviewed separately.
- The source dataset provides toxicity grade through AETOXGR but does not include AESEV, so the report describes toxicity grade rather than standard severity categories.

8. Conclusion

Milestone 5 integrates the portfolio's earlier source-data review, cross-domain validation, and subject-level analysis work into a complete clinical safety programming case study. Starting from SDTM AE and the Milestone 4 ADSL-style dataset, the workflow creates an ADAE-style analysis dataset, applies TEAE logic, produces incidence and grade summaries, creates a visualization, and validates the final outputs.

The analysis found that 420 of 453 safety subjects (92.7%) experienced at least one TEAE. Maximum Grade 3 or Grade 4 TEAEs occurred in 205 subjects (45.3%), and 138 subjects (30.5%) experienced at least one serious TEAE. Beyond these results, the primary portfolio value is the demonstrated ability to make defensible analysis decisions and implement them through modular, reproducible R programs with explicit quality control.

Appendix A. Program and Output Inventory

Programs

1. 00_setup.R - defines paths, loads packages, and creates milestone-specific directories.
2. 01_ae_source_assessment.R - assesses AE structure, terminology, timing, and clinical variables.
3. 02_build_adae.R - creates the ADAE-style dataset and treatment-period flags.
4. 03_create_teae_summary.R - creates overall, SOC, and PT TEAE summaries.
5. 04_create_teae_figure.R - creates the top-15 PT incidence figure.
6. 05_create_teae_grade_summary.R - creates maximum-grade and seriousness summaries.

Primary Deliverables

Type	File
Derived dataset	milestone5_adae.csv

Type	File
Analysis table	teae_summary_analysis.csv
Display table	teae_summary_table.csv
Analysis table	teae_grade_summary_analysis.csv
Display table	teae_grade_summary_table.csv
Figure	top_15_tae_incidence.png

Additional listings document source assessment, ADAE flag distributions, non-treatment-emergent and unclassified records, toxicity grade, seriousness, relationship, action taken, outcome, timing, event pattern, and source treatment-period information.